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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

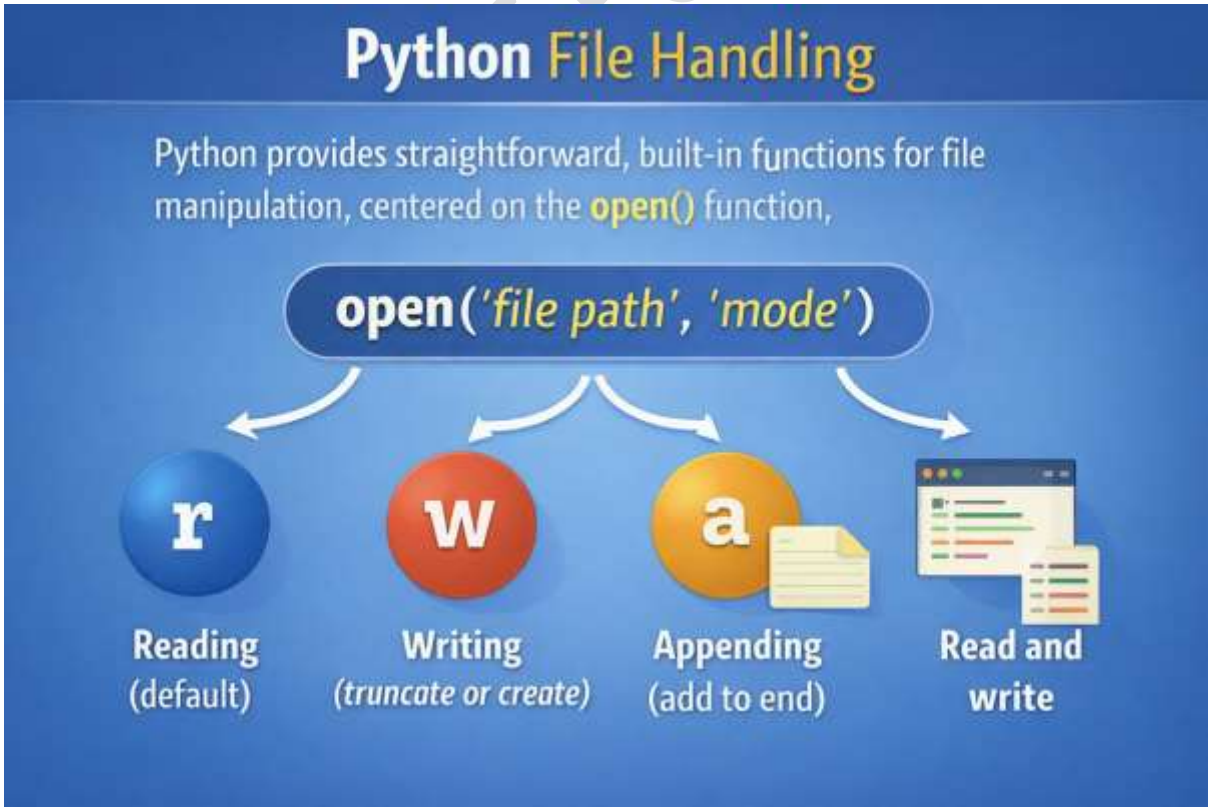
Classes : Offline & Online

Daily Teaching : 2 Hours

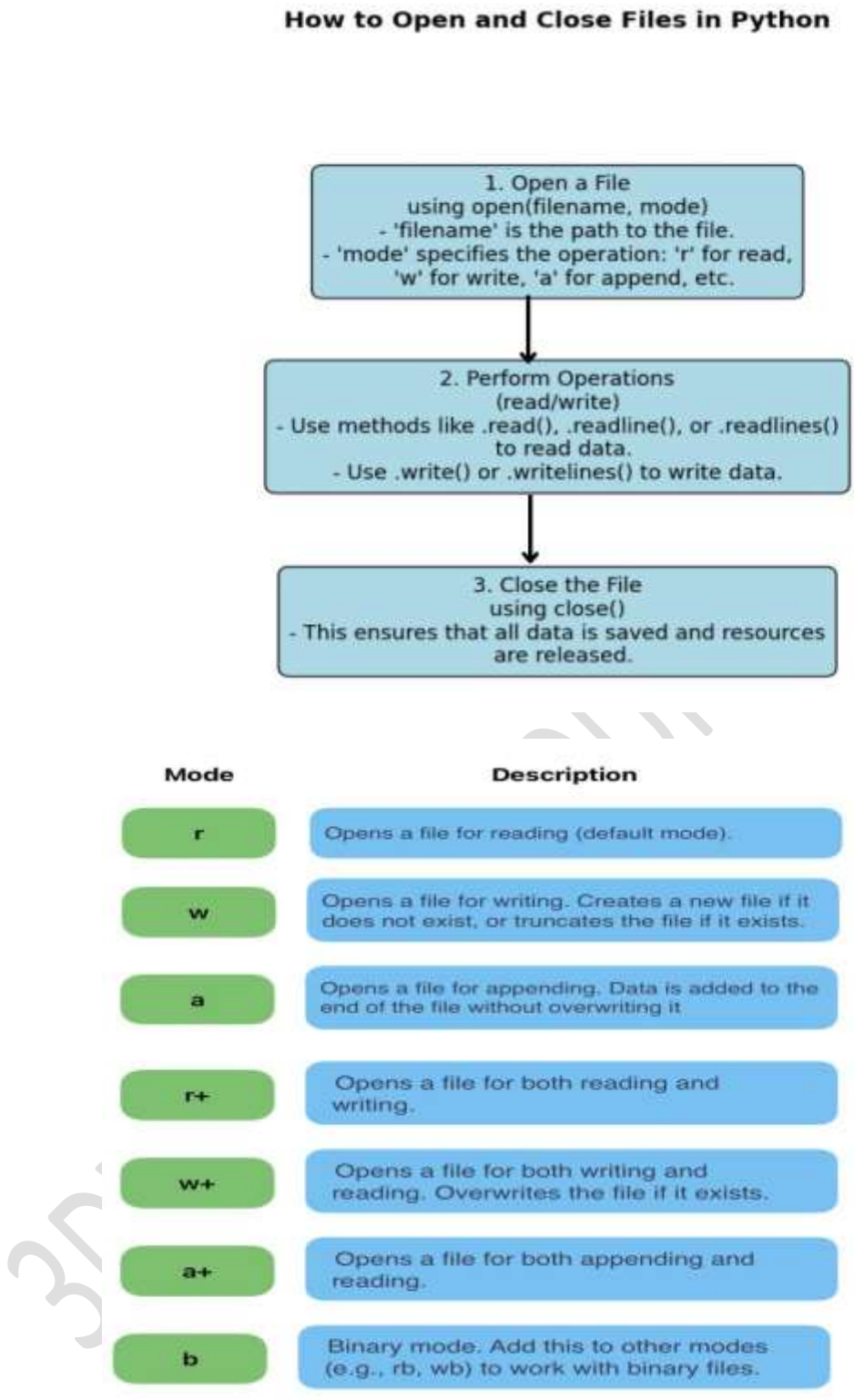
Total Classes : 100

Level 4 – AI Professional Foundation(11th–12th Standard)

Class 14 – File Handling in Python



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1. Learning Objectives

By the end of this lesson, students will be able to:

- Understand what file handling means.
- Explain why files are needed.
- Open and close files using Python.
- Read data from text files.
- Write data into text files.
- Append new data to existing files.
- Understand different file modes.
- Use with open() safely.
- Work with read(), readline(), and readlines().
- Create practical file-based programs.

2. What is File Handling?

File handling means using a program to create, read, write, update, and manage files stored on a computer.

For example, a student program may need to save:

Student Names

Marks

Attendance

Fees

Contact Details

If we only store the information in a Python variable:

students = ["Asha", "Rahul", "Priya"]

the information normally exists only while the program is running.

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A file allows us to **store data permanently** so it can be used later.

Simple idea

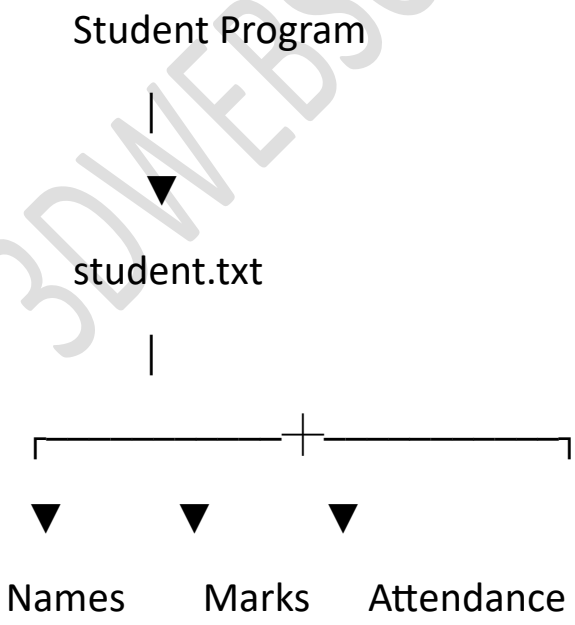


3. Why Do We Need Files?

Suppose a school has 500 students.

If the program closes, we don't want to lose all the information.

Instead:



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The next time the program starts, it can read the information from the file.

4. Types of Files

For beginners, files can be divided into two broad categories:

1. Text Files

Contain readable characters.

Examples:

students.txt

marks.txt

notes.txt

data.csv

2. Binary Files

Contain data stored in binary form.

Examples:

.jpg

.png

.mp3

.mp4

.pdf

.exe

In this lesson, we mainly focus on **text files**.

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5. What is a Text File?

A text file contains characters that humans can generally read.

Example:

students.txt

Asha

Rahul

Priya

Arjun

Python can read this information and process it.

6. File Handling Process

The basic file-handling process is:

OPEN

↓

READ / WRITE

↓

CLOSE

In Python:

```
file = open("students.txt", "r")

# perform operation

file.close()
```

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However, Python provides a safer and easier method using with open().

7. Opening a File

Python uses the open() function.

Syntax

open(filename, mode)

Example:

file = open("students.txt", "r")

Here:

students.txt → File name

r → Read mode

8. File Modes

File mode tells Python **what we want to do with the file**.

Mode	Meaning
r	Read
w	Write
a	Append
x	Create a new file
r+	Read and write

For this lesson, the most important modes are:

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r → Read

w → Write

a → Append

9. Read Mode – r

The r mode is used to read an existing file.

Suppose students.txt contains:

Asha

Rahul

Priya

Python program:

```
file = open("students.txt", "r")

data = file.read()

print(data)

file.close()
```

Output

Asha

Rahul

Priya

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10. read() Method

The read() method reads the contents of a file.

```
file = open("students.txt", "r")


data = file.read()


print(data)


file.close()
```

Flow

students.txt

|

| read()

▼

Python Program

|

▼

Display Data

11. Using with open()

A better approach is:

```
with open("students.txt", "r") as file:
   data = file.read()
```

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```
    print(data)
```

Python automatically handles closing the file when the with block finishes.

Recommended style

```
with open("students.txt", "r") as file:
    print(file.read())
```

12. Writing to a File – w

The w mode is used to write data into a file.

```
with open("students.txt", "w") as file:
    file.write("Asha\n")
    file.write("Rahul\n")
    file.write("Priya\n")
```

The file will contain:

Asha

Rahul

Priya

13. Important Warning About w Mode

The w mode can **overwrite existing content**.

Suppose the file initially contains:

Asha

Rahul

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Priya

Now:

```
with open("students.txt", "w") as file:
    file.write("Arjun")
```

The old content is replaced.

The file now contains:

Arjun

Remember:

w = Write/overwrite

14. New Line Character \n

The special character:

\n

means **new line**.

Example:

```
with open("students.txt", "w") as file:
    file.write("Asha\n")
    file.write("Rahul\n")
    file.write("Priya\n")
```

Output inside the file:

Asha

Rahul

Priya

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Without \n:

file.write("Asha")

file.write("Rahul")

the result would be:

AshaRahul

15. Writing Multiple Lines

We can use writelines().

students = ["Asha\n", "Rahul\n", "Priya\n"]

with open("students.txt", "w") as file:

    file.writelines(students)

The file contains:

Asha

Rahul

Priya

Important

writelines() does **not automatically add new lines**, so we include \n ourselves.

16. Append Mode – a

The a mode is used to add new data to the **end of an existing file**.

Suppose:

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students.txt

Asha

Rahul

Priya

Now:

```
with open("students.txt", "a") as file:
    file.write("Arjun\n")
```

The result becomes:

Asha

Rahul

Priya

Arjun

Remember:

a = Add to existing content

17. w vs a

w	**a**
Writes data	Adds data
Can overwrite old content	Preserves old content
Starts writing from beginning	Writes at the end
Use carefully	Useful for adding records

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Visual memory

w mode:

OLD DATA

↓

REPLACED

↓

NEW DATA

a mode:

OLD DATA

↓

KEPT

+

NEW DATA

18. Reading One Line – readline()

readline() reads one line at a time.

Suppose:

Asha

Rahul

Priya

Program:

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```
with open("students.txt", "r") as file:
    line = file.readline()
    print(line)
```

Output:

Asha

Calling readline() again reads the next line.

19. Reading All Lines – readlines()

readlines() reads all lines and returns them as a list.

```
with open("students.txt", "r") as file:
    lines = file.readlines()

print(lines)
```

Output:

['Asha\n', 'Rahul\n', 'Priya\n']

20. read() vs readline() vs readlines()

Method	Purpose
read()	Reads the entire file
readline()	Reads one line
readlines()	Reads all lines as a list

Easy memory:

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read() → Everything

readline() → One line

readlines() → List of lines

21. Reading a File Line by Line

We can also use a for loop.

```
with open("students.txt", "r") as file:
    for line in file:
        print(line.strip())
```

Why strip()?

A line read from a text file may contain \n.

strip() removes unnecessary spaces and newline characters from the beginning and end.

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Practical Session

22. Practical Program – Student Names

Step 1 – Create a file

Create:

students.txt

Add:

Asha

Rahul

Priya

Arjun

Step 2 – Read it using Python

```
with open("students.txt", "r") as file:
    for student in file:
        print(student.strip())
```

Output

Asha

Rahul

Priya

Arjun

23. Practical Program – Student Marks

Create:

marks.txt

Content:

85

72

90

65

78

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Python:

```
with open("marks.txt", "r") as file:
    for line in file:
        mark = int(line.strip())
        print(mark)
```

24. Calculate Total from a File

```
total = 0


with open("marks.txt", "r") as file:
    for line in file:
        mark = int(line.strip())
        total += mark


print("Total:", total)
```

Output

Total: 390

25. Calculate Average

```
total = 0
count = 0
```

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```
with open("marks.txt", "r") as file:
    for line in file:
        mark = int(line.strip())
        total += mark
        count += 1

average = total / count

print("Total:", total)
print("Average:", average)
```

Output

```
Total: 390
Average: 78.0
```

26. Writing Student Information

We can save student information to a file.

```
name = input("Enter student name: ")
age = input("Enter age: ")
course = input("Enter course: ")

with open("student_details.txt", "w") as file:
    file.write("Name: " + name + "\n")
    file.write("Age: " + age + "\n")
```

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```
    file.write("Course: " + course + "\n")

print("Student details saved successfully.")
```

Example input:

Enter student name: Asha

Enter age: 17

Enter course: Python

The file will contain:

Name: Asha

Age: 17

Course: Python

27. Append Multiple Students

```
name = input("Enter student name: ")
marks = input("Enter marks: ")

with open("students.txt", "a") as file:
    file.write(name + " - " + marks + "\n")

print("Student added successfully.")
```

Run the program multiple times.

The file could become:

Asha - 85

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Rahul - 72

Priya - 90

28. Checking Whether a File Exists

Python provides the os module.

```
import os


if os.path.exists("students.txt"):
    print("File exists")
else:
    print("File does not exist")
```

This is useful before trying to read a file.

29. Creating a New File – x

The x mode creates a new file.

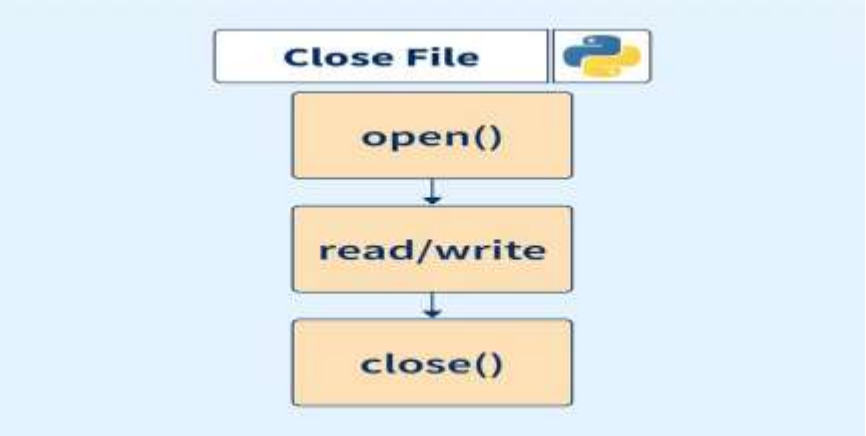
```
with open("newfile.txt", "x") as file:
    file.write("Hello Python")
```

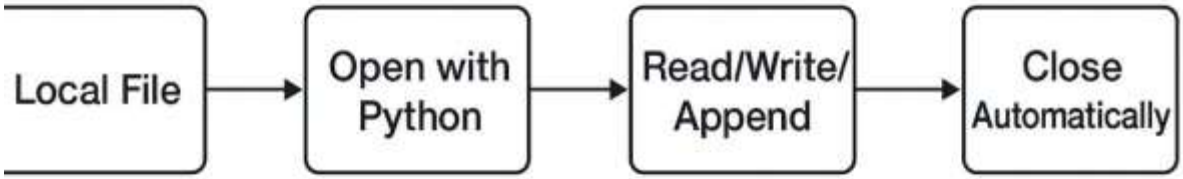
If the file already exists, Python raises an error.

This mode is useful when we specifically want to create a new file.

30. File Handling Flowchart

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Close File
open()
read/write
close()


Local File
Open with Python
Read/Write/ Append
Close Automatically



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START

|

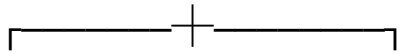
▼

Choose File

|

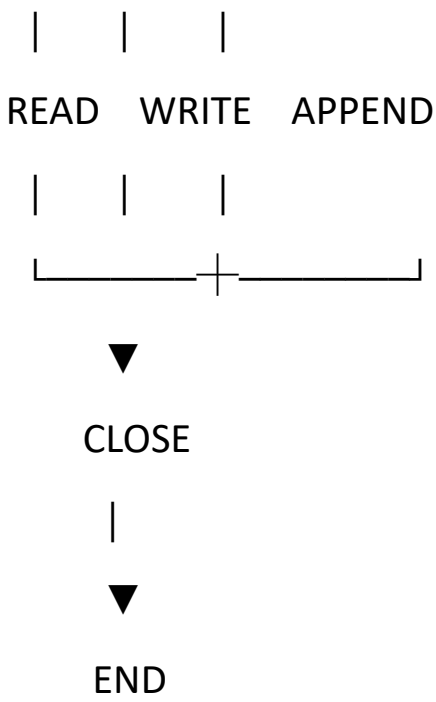
▼

Choose Mode



▼ ▼ ▼

r w a



31. File Handling with try-except

Sometimes the file may not exist.

Example:

```
try:
    with open("students.txt", "r") as file:
        print(file.read())

except FileNotFoundError:
    print("File not found.")
```

If the file does not exist:

File not found.

This prevents the program from stopping unexpectedly.

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32. Practical Project – Student Record File

Objective

Create a simple program that stores student names and marks in a text file.

```
def add_student():
    name = input("Enter student name: ")
    marks = input("Enter marks: ")

    with open("students.txt", "a") as file:
        file.write(name + "," + marks + "\n")

    print("Student added successfully.")



def view_students():
    try:
        with open("students.txt", "r") as file:
            for line in file:
                print(line.strip())

    except FileNotFoundError:
        print("No student records found.")
```

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add_student()

view_students()

Example Output

Enter student name: Asha

Enter marks: 85

Student added successfully.

Asha,85

33. Mini Project – Student Marks Manager

Create:

StudentMarks/

|

├— main.py

└— marks.txt

The program should provide:

===== STUDENT MARKS MANAGER =====

1. Add Student

2. View Students

3. Add More Records

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4. Exit

Sample Program

```
def add_student():
    name = input("Enter student name: ")
    marks = input("Enter marks: ")

    with open("marks.txt", "a") as file:
        file.write(name + "," + marks + "\n")

    print("Record saved.")
def view_students():
    try:
        with open("marks.txt", "r") as file:
            print("\nStudent Records")

            for line in file:
                name, marks = line.strip().split(",")
                print("Name:", name, "| Marks:", marks)
    except FileNotFoundError:
        print("No records available.")
while True:
    print("\n===== STUDENT MARKS MANAGER =====")
    print("1. Add Student")
```

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```
    print("2. View Students")
    print("3. Exit")
    choice = input("Enter choice: ")
    if choice == "1":
        add_student()
    elif choice == "2":
        view_students()
    elif choice == "3":
        print("Thank you!")
        break

    else:
        print("Invalid choice.")
```

34. Expected Output

```
===== STUDENT MARKS MANAGER =====
1. Add Student
2. View Students
3. Exit

Enter choice: 1

Enter student name: Rahul
```

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Enter marks: 78

Record saved.

Later:

Enter choice: 2

Student Records

Name: Rahul | Marks: 78

Name: Asha | Marks: 85

Name: Priya | Marks: 92

35. Important File Methods

Method	Purpose
open()	Opens a file
read()	Reads complete content
readline()	Reads one line
readlines()	Reads all lines as a list
write()	Writes text
writelines()	Writes multiple strings
close()	Closes the file
strip()	Removes surrounding whitespace

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36. Important File Modes

Mode	Purpose
r	Read
w	Write / overwrite
a	Append
x	Create new file
r+	Read and write

Student Memory Trick

RWA

R → Read

W → Write

A → Add

37. Common Mistakes

Mistake 1 – Wrong file name

open("student.txt", "r")

but the actual file is:

students.txt

This causes a file-not-found error.

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Mistake 2 – Forgetting \n

file.write("Asha")

file.write("Rahul")

Output:

AshaRahul

Use:

file.write("Asha\n")

file.write("Rahul\n")

Mistake 3 – Accidentally overwriting data

open("students.txt", "w")

may replace existing content.

If you want to add a new record:

open("students.txt", "a")

Mistake 4 – Forgetting to convert numbers

File contents are read as text.

For example:

mark = file.readline()

mark is a string.

To perform numerical calculations:

mark = int(mark)

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38. Real-World Applications

File handling is used in many applications:

School Management

↓

Student Records

Banking

↓

Transaction Records

Hospital

↓

Patient Records

Shopping

↓

Product Information

Attendance

↓

Attendance Records

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Log Systems

↓

Activity Logs

39. Practical Classroom Activities

Activity 1 – Personal Diary

Create:

diary.txt

Ask the user:

What did you learn today?

Save the answer using append mode.

Activity 2 – Number File

Create a file containing:

10

20

30

40

50

Write a Python program to:

- Read the numbers
- Find the total
- Find the average

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- Find the largest number

Activity 3 – Student Records

Create a program that stores:

Student Name

Roll Number

Marks

in a text file.

40. Mini Challenge

Ask students to create a **To-Do List Application**.

Requirements

===== TO-DO LIST =====

1. Add Task

2. View Tasks

3. Exit

Tasks should be saved permanently in:

tasks.txt

Example

1. Complete Python homework

2. Practice file handling

3. Prepare project

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Students should use:

- open()
- with
- write()
- read()
- append
- for loop

41. Quick Revision

What is file handling?

Working with files to store and retrieve data.

What is open()?

Used to open a file.

What is r?

Read mode.

What is w?

Write mode; it can overwrite existing content.

What is a?

Append mode; adds content to the end.

What does read() do?

Reads the complete file.

What does readline() do?

Reads one line.

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What does readlines() do?

Returns all lines as a list.

Why use with open()?

It provides a convenient way to work with a file and ensures the file is properly closed.

42. Knowledge Check

1. What is file handling?

2. Write the syntax of open().

3. What is the difference between w and a?

4. What does \n represent?

5. What is the purpose of read()?

6. What is the purpose of readline()?

7. What is the purpose of readlines()?

8. Why is with open() recommended?

9. Which mode is used to append data?

10. Which mode can overwrite existing data?

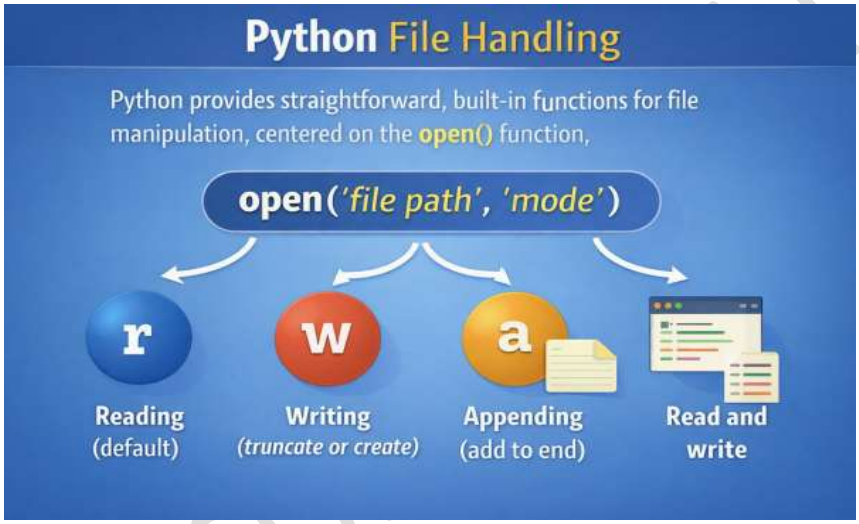
Answers

1. Managing files through a program.
2. open(filename, mode)
3. w writes/overwrites; a adds to the end.
4. New line.
5. Reads the entire file.

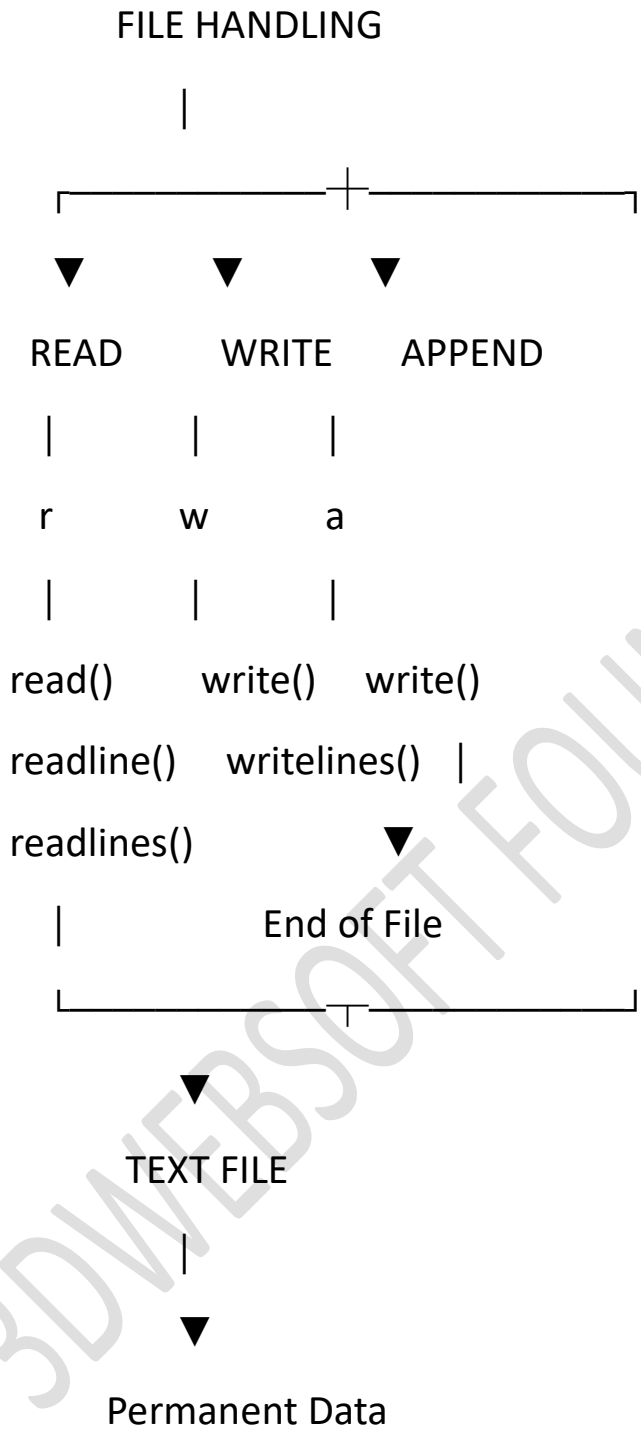
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6. Reads one line.
7. Reads all lines into a list.
8. It manages the file context and closes the file automatically.
9. a
10. w

43. Final Concept Map







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Student Takeaway

File handling allows Python programs to store data permanently. The three most important operations are r for reading, w for writing, and

a for appending. For safe and clean programs, students should become comfortable with with open(...) as file:.

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